

Lab 1: Linear Regression. Predicting asking prices for used Corollas

70-467 - Machine Learning for Business Analytics

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A Toyota used-car operation in the Netherlands posts asking prices on Corollas that are on sale in late summer 2004. When a car arrives, the spec sheet lists age, kilometres, fuel, and options. The manager has to set a price. If the ask is high, the car sits. If it is low, the dealership forgoes euros a buyer would have paid. The number also has to be something they can explain if a buyer asks why this Corolla is priced that way.

The manager has 1,436 asking prices from that season, in euros. Each listing records age in months as of August 2004, kilometres, fuel type, engine, and a long option list of other car features. Those listings are the comparables. They are asking prices, not the dealer's cost and not confirmed sales. The codebook on the next page gives type and unit for every column.

Staff do not use the comparables in one way. A common rule is age: cars of the same age get similar prices. Some people also look at the odometer (KMs traveled), or at diesel versus petrol. Some treat automatic climate control as a different product and would not set the price from age alone. If those differences are large in euros, leaving them out will change the posted price. If they are small next to how much listings already scatter, a simpler rule may be enough.

Your assignment is to produce an asking price for a car with a given specification, and to say how wide a band is honest for that one car. Coefficients along the way are how you build and defend the guess: which specs move euros, holding the other recorded features constant. What can the dealer use this information for? You will price two cars.

Car A is already listed. It is row 1 in the file: a diesel TERRA hatchback, 23 months old, 46,986 km, no automatic climate control, asking 13,500 euros. The model can be checked against a number that is already on the lot.

Car B is not in the file. Petrol, 36 months old, 60,000 km, with automatic climate control. There is no listed price to look up. You have to price it from the comparables.

The file does not contain the dealership's cost, the cost of fitting an option, or how many extra buyers would appear if the ask were cut by 500 euros. Estimates from this file are associations among asking prices in 2004. They do not measure the effect of changing a price, and they are not a 2026 valuation.

Read the codebook before class. In the room you will fit the comparables in R, then post a number and a band for Car A and for Car B. After class you will have additional questions to solve and understand also connected to our regression slides.

Codebook. 1,436 rows, 39 columns. Dummy means 0 = no and 1 = yes.

Variable	Type	Unit	Description
Id	identifier	none	Row identifier. Not a feature of the car.
Model	text	none	Full string: trim, body, doors.
Price	numeric	euros	Asking price posted on the listing.
Age_08_04	numeric	months	Age of the car as of August 2004.
Mfg_Month	numeric	1 to 12	Month of manufacture.
Mfg_Year	numeric	year	Year of manufacture. Nearly the same information as age.
KM	numeric	kilometres	Odometer reading.
Fuel_Type	factor	none	Petrol, Diesel, or CNG.
HP	numeric	horsepower	Engine power.
Met_Color	dummy	0 or 1	Metallic paint.
Color	factor	none	Paint colour.
Automatic	dummy	0 or 1	Automatic transmission.
CC	numeric	cubic cm	Engine displacement. One row is 16000 (Id 81): a 1.6 litre with an extra zero.
Doors	numeric	count	Number of doors.
Cylinders	numeric	count	Number of cylinders. Always 4 in this file.
Gears	numeric	count	Number of gears.
Quarterly_Tax	numeric	euros	Quarterly road tax (Netherlands).
Weight	numeric	kilograms	Vehicle weight.
Mfr_Guarantee	dummy	0 or 1	Still covered by the manufacturer guarantee.
BOVAG_Guarantee	dummy	0 or 1	Covered by a BOVAG (Dutch dealer network) guarantee.
Guarantee_Period	numeric	months	Guarantee length.
ABS	dummy	0 or 1	Anti-lock brakes.
Airbag_1	dummy	0 or 1	Driver airbag.
Airbag_2	dummy	0 or 1	Passenger airbag.
Airco	dummy	0 or 1	Air conditioning.
Automatic_airco	dummy	0 or 1	Automatic climate control.
Boardcomputer	dummy	0 or 1	Onboard computer.
CD_Player	dummy	0 or 1	CD player.
Central_Lock	dummy	0 or 1	Central locking.
Powered_Windows	dummy	0 or 1	Powered windows.
Power_Steering	dummy	0 or 1	Power steering.
Radio	dummy	0 or 1	Radio.
Mistlamps	dummy	0 or 1	Mist lamps.
Sport_Model	dummy	0 or 1	Sport model.
Backseat_Divider	dummy	0 or 1	Backseat divider.
Metallic_Rim	dummy	0 or 1	Metallic rims.
Radio_cassette	dummy	0 or 1	Radio cassette.
Parking_Assistant	dummy	0 or 1	Parking assistant. Four cars in the file.
Tow_Bar	dummy	0 or 1	Tow bar.